Towards a Linear-Algebraic Hypervisor

Breandan Mark Considine

June 10, 2026

The RASP Model

Definition (Cook & Reckhow, '73)

A RASP machine, $\mathcal{R} = \langle C, c_0, \to, F \rangle$ consists of a configuration space, $\langle i, a, M, u, y \rangle : C = \mathbb{N} \times \mathbb{Z} \times \mathbb{Z}^{\mathbb{N}} \times \mathbb{Z}^* \times \mathbb{Z}^*$. A RAS program is a vector $P : (\mathbb{Z} \times \mathbb{Z})^m$ packed with an *input* $x : \mathbb{Z}^*$, into an *initial configuration*, $c_0 = \left\langle 0, 0, M'_{0..2m} \leftarrow P, x0, \varepsilon \right\rangle$. Let $\langle o, j \rangle = \langle M_i, M_{i+1} \rangle$ denote the *opcode* and *operand* and $\to$ be defined as:

$$\langle i, a, M, u, y \rangle \to \begin{cases} \langle i+2 \,,\, j \,,\, M \,,\, u \,,\, y \rangle, & \text{if } o = 1_{\texttt{LOD}}, \\ \langle i+2 \,,\, a + M_j \,,\, M \,,\, u \,,\, y \rangle, & \text{if } o = 2_{\texttt{ADD}}, \\ \langle i+2 \,,\, a - M_j \,,\, M \,,\, u \,,\, y \rangle, & \text{if } o = 3_{\texttt{SUB}}, \\ \langle i+2 \,,\, a \,,\, M_j \leftarrow a \,,\, u \,,\, y \rangle, & \text{if } o = 4_{\texttt{STO}}, \\ \langle \;\; j \;\; \,,\, a \,,\, M \,,\, u \,,\, y \rangle, & \text{if } o = 5_{\texttt{BPA}} \;\wedge\; a > 0, \\ \langle i+2 \,,\, a \,,\, M \,,\, u \,,\, y \rangle, & \text{if } o = 5_{\texttt{BPA}} \;\wedge\; a \leq 0, \\ \langle i+2 \,,\, a \,,\, M_j \leftarrow \sigma \,,\, v \,,\, y \rangle, & \text{if } o = 6_{\texttt{RD}} \;\wedge\; u = \sigma v, \\ \langle i+2 \,,\, a \,,\, M \,,\, u \,,\, y \cdot M_j \rangle, & \text{if } o = 7_{\texttt{PRI}}, \\ \langle \;\; i \;\; \,,\, a \,,\, M \,,\, u \,,\, y \rangle, & \text{otherwise.} \end{cases}$$

and the halting configurations are simply $F = \{c \in C \mid c \to c\}$.

W-RASP: A Bounded Variant

Fix $w, m, n, \ell, s \in \mathbb{N}$, assume $\mathbb{W} = \mathbb{F}_2^w$, and for $P : \mathbb{W}^{2m \le n}$ define:

$$\mathcal{R}^{(w,n,\ell,s)} = \langle C_w, c_0, \to, F \rangle, \quad C_w = \mathbb{W} \times \mathbb{W} \times \mathbb{W}^n \times \mathbb{W}^{\ell+1} \times \mathbb{W}^{s+1}.$$

The transition operator takes a configuration $\langle i, a, M, u, y \rangle : C_w$ to:

$$\begin{cases}
\langle i \oplus 2_w \,, j \,, M \,, u \,, y \rangle, & \text{if } o = 1_{\texttt{LOD}}, \\
\langle i \oplus 2_w \,, a \oplus M_{j \bmod n} \,, M \,, u \,, y \rangle, & \text{if } o = 2_{\texttt{ADD}}, \\
\langle i \oplus 2_w \,, a \otimes M_{j \bmod n} \,, M \,, u \,, y \rangle, & \text{if } o = 3_{\texttt{MUL}}, \\
\langle i \oplus 2_w \,, a \,, M_{j \bmod n} \leftarrow a \,, u \,, y \rangle, & \text{if } o = 4_{\texttt{STO}}, \\
\langle \quad j \quad \,, a \,, M \,, u \,, y \rangle, & \text{if } o = 5_{\texttt{BNZ}} \ \wedge \ a \neq 0, \\
\langle i \oplus 2_w \,, a \,, M \,, u \,, y \rangle, & \text{if } o = 5_{\texttt{BNZ}} \ \wedge \ a = 0, \\
\langle i \oplus 2_w \,, a \,, M_{j \bmod n} \leftarrow u^{\bullet} \,, u^{\triangleright} \,, y \rangle, & \text{if } o = 6_{\texttt{RD}} \ \wedge \ u_0 < \ell, \\
\langle i \oplus 2_w \,, a \,, M \,, u \,, y :: M_{j \bmod n} \rangle, & \text{if } o = 7_{\texttt{PRI}}, \\
\langle \quad i \quad \,, a \,, M \,, u \,, y \rangle, & \text{otherwise.}
\end{cases}$$

$$\langle o, j \rangle := \left\langle M_{i \bmod n}, M_{(i \oplus 1) \bmod n} \right\rangle, \qquad M_j \leftarrow a := \left[\delta_{jk} a + (1 - \delta_{jk}) M_k \right]_{k=0}^{n-1}.$$

$$u^{\bullet} := u_{\min(u_0+1,\ell)}, \qquad u^{\triangleright} := u_0 \leftarrow \min(u_0 + 1, \ell),$$

$$y :: z = \begin{cases} (y_{y_0+1} \leftarrow z)_0 \leftarrow y_0 + 1, & y_0 < s, \\ y, & \text{otherwise.} \end{cases}$$

Heapless Array Language

We define a simple array langauge for functions $f : \mathbb{N}^{\lambda \leq \ell} \rightarrow \mathbb{N}^{\eta \leq \sigma}$:

```
F ::= fun f0 ( ipt :  W ^ N ) -> W ^ N { B }
N ::= 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9
B ::= S | S ; B | hlt | brk
S ::= P = E | ife G { B } { B } | whl G { B }
Z ::= 0 | N
G ::= ipt [ Z ] | scr [ Z ] | opt [ Z ]
P ::= scr [ Z ] | opt [ Z ]
E ::= G + G | G * G | G + Z | G * Z | G | Z
```

`ipt`, `opt` and `scr` refer to the *input*, *output* and *scratch* buffers, respectively. Permissions are RO for `ipt` and RW for `opt`, `scr`. `hlt` returns immediately and `brk` escapes the local scope.

Heapless Array Lowering

$$\frac{\Gamma[\kappa \mapsto \ell_f] \vdash B \Rightarrow C}{\vdash \texttt{fun f0 (ipt : I ∧ } \lambda \texttt{) -> I ∧ } \eta \texttt{ \{ } B \texttt{ \} } \Rightarrow C\ {}^{\ell_f:}\boxed{\texttt{PRI } q_i}_{i=0}^{\eta-1}\boxed{\texttt{HLT}}}\ \textsc{Fun}$$

$$\frac{}{\Gamma \vdash \alpha[z] \Rightarrow_a \boxed{\texttt{LOD 0}}\boxed{\texttt{ADD } [\![\alpha]\!]_z}}\ \textsc{Get} \qquad \frac{\Gamma \vdash E \Rightarrow_a A}{\Gamma \vdash \alpha[z] = E \Rightarrow A\boxed{\texttt{STO } [\![\alpha]\!]_z}}\ \textsc{Put}$$

$$\frac{\Gamma \vdash I_1 \Rightarrow_a J_1 \qquad \Gamma \vdash I_2 \Rightarrow_a J_2}{\Gamma \vdash I_1 + I_2 \Rightarrow_a J_1\boxed{\texttt{STO } \nu}J_2\boxed{\texttt{ADD } \nu}}\ \oplus \qquad \frac{\Gamma \vdash I_1 \Rightarrow_a J_1 \qquad \Gamma \vdash I_2 \Rightarrow_a J_2}{\Gamma \vdash I_1 \times I_2 \Rightarrow_a J_1\boxed{\texttt{STO } \nu}J_2\boxed{\texttt{MUL } \nu}}\ \otimes$$

$$\frac{\Gamma \vdash H \Rightarrow_a J \qquad \Gamma \vdash B_1 \Rightarrow C_1 \qquad \Gamma \vdash B_2 \Rightarrow C_2}{\Gamma \vdash \texttt{ife } H \texttt{ \{}B_1\texttt{\} \{}B_2\texttt{\}} \Rightarrow J\boxed{\texttt{BNZ } \ell_t}C_2\boxed{\texttt{JMP } \ell_e}\ {}^{\ell_t:}C_1\ {}^{\ell_e:}}\ \textsc{Ife}$$

$$\frac{\Gamma \vdash H \Rightarrow_a J \qquad \Gamma[\kappa \mapsto \ell_e] \vdash B \Rightarrow C}{\Gamma \vdash \texttt{whl } H \texttt{ \{}B\texttt{\}} \Rightarrow {}^{\ell_h:}J\boxed{\texttt{BNZ } \ell_b}\boxed{\texttt{JMP } \ell_e}\ {}^{\ell_b:}C\boxed{\texttt{JMP } \ell_h}\ {}^{\ell_e:}}\ \textsc{While} \qquad \frac{\Gamma(\kappa) = \ell}{\Gamma \vdash \texttt{brk} \Rightarrow \boxed{\texttt{JMP } \ell}}\ \textsc{Break}$$

$$\frac{\Gamma \vdash B \Rightarrow C_B \qquad \Gamma \vdash S \Rightarrow C_S}{\Gamma \vdash B \ ; \ S \Rightarrow C_B\ C_S}\ \textsc{Seq} \qquad \frac{}{\Gamma \vdash \texttt{hlt} \Rightarrow \boxed{\texttt{PRI } q_i}_{i=0}^{l-1}\boxed{\texttt{HLT}}}\ \textsc{Halt}$$

We write $\underline{\Gamma \vdash E \Rightarrow A}$ when the A implements E, and $\underline{\Gamma \vdash E \Rightarrow_a A}$ when a is left in the accumulator. Desugar $\boxed{\texttt{JMP } \ell}$ as $\boxed{\texttt{LOD 1}}\boxed{\texttt{BNZ } \ell}$ and interpret array indexing modulo declared length, i.e., $[\![\alpha]\!]_z := M_{2m+[\![\alpha]\!]+(z \bmod |\alpha|)}$.

Memory Layout

Reserve disjoint memory regions $p_{0\ldots\lambda-1}$, $q_{0\ldots\eta-1}$, and $r_{0\ldots\gamma-1}$ for arrays $\texttt{ipt} : \mathbb{W}^{\lambda\le\ell}$, $\texttt{opt} : \mathbb{W}^{\eta\le\sigma}$ and $\texttt{scr} : \mathbb{W}^{\gamma\le\mu}$, with $(5+\ell+\sigma+\mu+2m) \le n$. The hypervisor stores each VM as one contiguous 32-bit row; RASP instructions address only M_v.

$$\mathrm{VM}_v = \overbrace{[\texttt{pc}, \texttt{acc}, \texttt{steps}, \texttt{halted}, \texttt{outCount}, \texttt{out}_0, \texttt{out}_1, \texttt{out}_2]}^{H_v\text{: hypervisor sidecar}} \;\|\; \overbrace{[M_v[0], \ldots, M_v[n-1]]}^{M_v\text{: RASP-addressable memory}}, \quad n \approx 2^8.$$

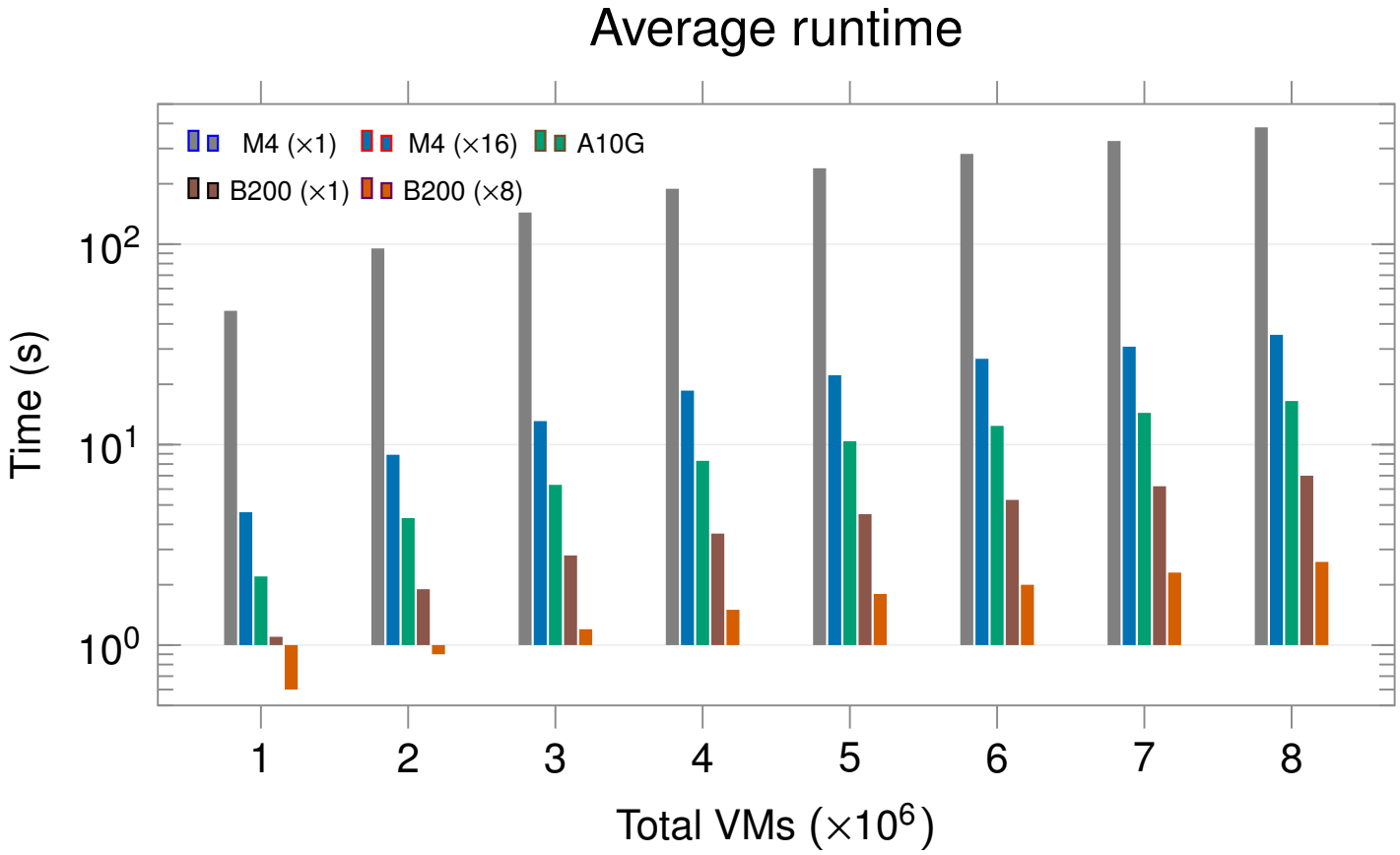
$$\texttt{MEM\_BASE} = |H_v| = 8, \qquad \mathrm{VM}_v[\texttt{MEM\_BASE} + a] = M_v[a].$$

$$M_v = \Big[\overbrace{\underset{\texttt{LOD 0}}{1}, \underset{\texttt{ADD } p_0}{66}, \underset{\texttt{STO } q_0}{76}, \underset{\texttt{LOD 1}}{9}, \underset{\texttt{BNZ 6}}{53}, \underset{\texttt{PRI } q_0}{79}, \underset{\texttt{HLT}}{0}, \ldots}^{P_{0..c-1}\text{ packed code ((arg}\ll 3)|\text{op)}}, \underbrace{0, \ldots, 0}_{\ell}, \overbrace{0, \ldots, 0}^{\texttt{ipt}=p}, \overbrace{0, \ldots, 0}^{\texttt{opt}=q}, \overbrace{0, \ldots, 0}^{\texttt{scr}=r}, \underset{\nu\text{ scratch}}{0}, \overbrace{0, \ldots, 0}^{\text{padding}}\Big]$$

$$p = c, \qquad q = c+\ell, \qquad r = c+\ell+s, \qquad \nu = c+\ell+s+\gamma, \qquad c+\ell+s+\gamma+1 \le n$$

Arithmetic keeps the low 29 bits; $\texttt{ipt/opt/scr}$ indices wrap modulo ℓ, s, γ.

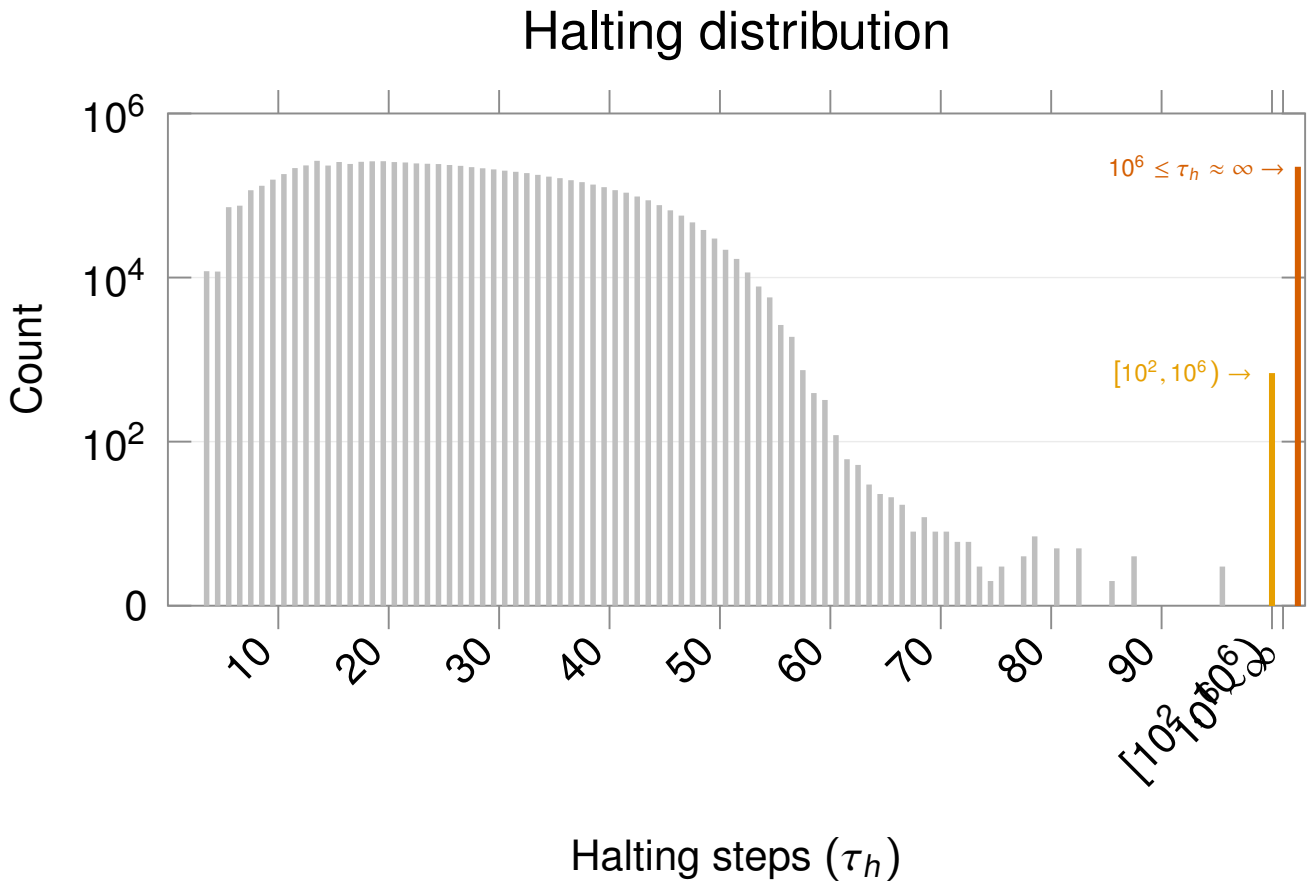
Evaluation: Runtime



$w = 32$, $n = 250$, $\ell = 10$, $s = 2$, $\tau_{\max} = 10^6$, $d = 8 \times 10^6$, $L = 100$.

Evaluation: Halting Distribution

$$\Pr_{P' \sim \mathcal{P}_L}\left(\Phi_w^{\tau_{\max}}\left(c_0(P' \Rightarrow P, x \sim \mathbb{F}_2^{w \cdot \mathrm{arity}(P')})\right) \in F \;\middle|\; L = |P'|\right).$$



We plot programs by their halting number τ_h, the least τ such that $\Phi_w^{\tau}(c_0) = \Phi_w^{\tau+1}(c_0)$.